KEYSTONE-MPP-F1 --- DRAFT PREAUTHORIZATION MANUSCRIPT

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Title

KEYSTONE: A Draft Study of Dispute-Key Serviceability Distinct from Ciphertext Availability in Encrypted AI Rollups

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Abstract

Encrypted optimistic AI rollups can preserve ciphertext availability while still failing to deliver timely dispute-key serviceability. KEYSTONE studies that separation through a non-production readiness-audit workflow assembled from established threshold-decryption components rather than a new cryptographic primitive. The present draft is bounded to three research-case claims: a property-separation hypothesis (`C001`), an internal prototype and reproducibility hypothesis (`C002`), and a broader solution-viability hypothesis that still requires external evidence (`C003`). Existing tests, simulations, figures, diagrams, and contract-boundary artifacts are internal, analytic, preliminary, or exploratory evidence only. Novelty remains unresolved, feasibility is unassessed in canonical state, and confirmatory execution is not authorized. Accordingly, this shell records admissible wording, planned sections, and explicit evidence gaps; it does not make a publication-readiness claim.

1. Introduction

The motivating problem is not ciphertext publication alone. A rollup dispute may require authorized recovery of a record key or threshold-decryption output

within a deadline, and that requirement can fail even when ciphertext bytes or receipts remain retrievable. The current draft treats that gap as the core research question.

The manuscript remains within FREEZE.md, docs/11_PAPER_BLUEPRINT.md, and the canonical novelty case. It does not claim a new threshold-decryption primitive, timing-free accountability, adaptive selective-withholding resistance, unconditional future availability, or production security.

The three manuscript-level claim anchors are:

- C001: study and formalize dispute-key availability as distinct from ciphertext availability under the frozen model, subject to unresolved strongest-prior-art closure;
- C002: document the internal, non-production prototype and local reproducibility boundary;
- C003: test the frozen analytic, simulated, prototype, and eventually external evidence package without wording above its current maturity ceiling.

Methods

The detailed methods architecture is organized across Sections 2–6. Those sections remain a pre-authorization manuscript view of the canonical novelty, protocol, analysis-plan, and implementation artifacts.

The feasibility package now includes a non-executable confirmatory pilot plan, machine-readable result contract, and a result-blind PC03 amendment. They reserve all nine RID-* result families, separate the minimum short-paper core from the extended deadline lane, prohibit reuse of historical outputs as confirmatory evidence, and leave all observed outcomes missing. For three minimum synthetic cells—IID baseline, stratified-versus-uniform at s=8, and selective withholding at w=11—the draft amendment predeclares 32 primary and four reserve seed blocks, 4,096 draws per block, result-blind precision rules, and deterministic seed schedules. Correlated-domain analysis is EXCLUDED_PENDING_TRUTHFUL_DOMAIN_LABEL_SOURCE; its current plots remain exploratory only. The distributed deadline family is EXCLUDED_PENDING_ENVIRONMENT_PROFILE. This design-readiness package does not issue GO: it still needs independent methods verification and separate execution authority.

2. Background and Related Work

PENDING - NOVELTY_UNRESOLVED. This section will compare KEYSTONE with data availability sampling, threshold encryption and verifiable partial decryption, VSS/PVSS and DKG, proactive or dynamic sharing, proof of retrievability, encrypted execution, threshold KMS, and accountable liveness. The current novelty matrix rejects broad primitive or first-system novelty. Until the independent, signed novelty closure exists, only narrow property-separation and

integration language is admissible.

PeerDAS is the relevant contrast class for public or ciphertext-oriented data availability rather than dispute-key serviceability (Ryan et al. 2024). Established secret-sharing, PVSS, proactive sharing, robust DPSS, threshold-decryption, and DLEQ-style proof lineage are adopted building blocks rather than claimable novelty (Shamir 1979; Schoenmakers 1999; Herzberg et al. 1995; Yurek et al. 2023; Boneh et al. 2006; Chaum and Pedersen 1993).

The adjacent encrypted-rollup, threshold-release, and authorized-decryption design space is already populated by encrypted optimistic-execution and threshold-release systems, including EigenAI, ETHTID, Ferveo, vetKeys, BEAT-MEV, practical one-time-setup batched threshold encryption, time-lock encrypted storage, context-dependent threshold decryption, and consensus- authorized threshold decryption for voting (Alves et al. 2026; Stengele et al. 2021; Bebel and Ojha 2022; Cerulli et al. 2023; Bormet et al. 2025; Choudhuri et al. 2025; Agarwal et al. 2025; Boneh et al. 2025; Liu et al. 2026). Accountability and timing assumptions are also prior-art surfaces rather than KEYSTONE-owned primitives (Boneh et al. 2023; Lewis-Pye et al. 2025). Accordingly, the only currently admissible contribution language is the narrower distinction between ciphertext availability and dispute-key serviceability under the frozen model, plus the claim-bounded audit and analysis surfaces that still require independent review.

3. Model and Definitions

The pre-authorization formal source defines ciphertext availability, valid-share registration, audit-time readiness, authorized decryptability, deadline liveness, pre-authorization confidentiality, and conditional dispute-key availability. Three constructive obligations are currently preserved: (i) ciphertext availability does not imply deadline-bounded dispute-key availability when only `t-1` record contributions are serviceable; (ii) a routine audit may pass while a target-aware selective withholder prevents an authorized dispute from reaching `t` contributions; and (iii) a finite audit prefix cannot guarantee unconditional future availability without temporal assumptions. These are draft logical non-implication witnesses, not prevalence, novelty, or production-security results. The paper does not claim a complete pairwise property lattice, and all three obligations remain subject to formal and novelty review.

4. KEYSTONE Protocol

The authoritative pre-authorization protocol is `research-case/03-design/protocol.md`. The manuscript-facing protocol will cover epoch setup, record KEM/DEM handling, beacon-derived canary challenges, context-bound partial decryptions and proofs, audit verification, authorized dispute release, evidence records, recovery, and domain-aware placement.

The current local package contains deterministic transcript and signature fixtures that exercise audit-request and partial-response interfaces. These are interoperability fixtures, not empirical evidence or proof of production security. The current local audit API accepts beacon and context inputs and derives the canary challenge internally, removing the direct caller-supplied group-element path from routine audit execution. This is a pre-authorization prototype boundary, not a formal non-substitution proof or production-security claim, and it still requires independent cryptographic review. The current draft treats threshold decryption, DLEQ-style verification, and hash-to-curve usage as adopted components rather than new cryptographic claims (Boneh et al. 2006, 2025; Chaum and Pedersen 1993; Faz-Hernandez et al. 2023).

5. Analysis

The current analysis surface includes a static catastrophic-state hypergeometric calculation and internal simulation models for IID, correlated-domain, stratified-sampling, and selective-withholding scenarios. Any static probability statement must be qualified as one-audit reasoning under the declared fixed-ready-set and sampling assumptions. Repeated-audit and deadline claims require explicit temporal and synchrony assumptions.

Current boundaries are mandatory:

- temporal Markov outputs are exploratory only;
- selective withholding is a negative finding and limitation;
- deadline success remains blocked pending an accountable environment profile, synchronized trace design, and authorized distributed benchmark;
- no analysis statement may exceed the claim ceilings encoded in the canonical research case.

6. Implementation

The workspace contains a Python threshold-KEM/DLEQ prototype, deterministic fixtures, local experiment scripts, and a Solidity bulletin-board boundary. Local cryptographic timings and contract test-body gas measurements are implementation artifacts only. They do not establish production throughput, deployment cost, audited cryptographic security, or independent reproducibility.

Two fail-closed pre-authorization engineering receipts currently bind local source/configuration hashes to observed test outcomes. The crypto receipt maps 22 pytest nodes and two vector-freshness checks; the contract receipt maps 27 Foundry cases, including 14 deterministic boundary tests, three fuzz properties, eight test-body gas operations, and two stateful invariants. All mapped cases passed in the current same-workspace runs. The receipts explicitly remain non-empirical, non-independent engineering QA; they are not canonical scientific results, cryptographic or smart-contract audits, or production validation. An editable pre-authorization QA cov-

erage table is generated from the two receipts and their hash sidecars at `paper/tables/preauthorization_engineering_qa.csv` for paper drafting, but it is not a canonical T-series result table and cannot be used to promote any scientific claim or raise the maturity or supported scope of `C002`.

Production DKG/PVSS, audited cryptographic libraries, distributed deployment, and independently controlled reproduction remain outside the current internal prototype boundary.

6.1 Draft protocol and method display map

The following diagram callouts bind the current editable sources and rendered derivatives to their intended manuscript roles. They communicate definitions, protocol structure, methods, and limitations; they are not result evidence and do not raise the maturity of `C001`, `C002`, or `C003`. Final placement remains subject to the target venue, final-size review, and accountable-human approval.

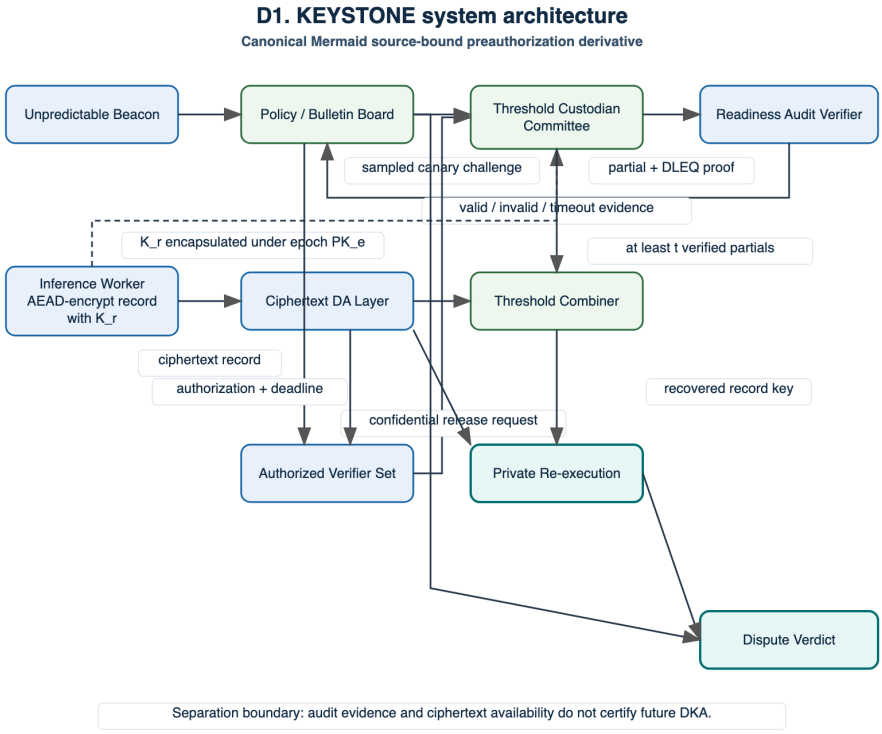


Figure 1: D1. System architecture separating ciphertext publication, policy coordination, readiness auditing, authorized release, threshold combination, private re-execution, and the dispute verdict.

D1 — System architecture. Ciphertext availability, routine canary auditing,

and authorized dispute-key release are shown as separate paths. The diagram is a protocol map for C002/C003, not evidence that each path is externally available or production-secure.

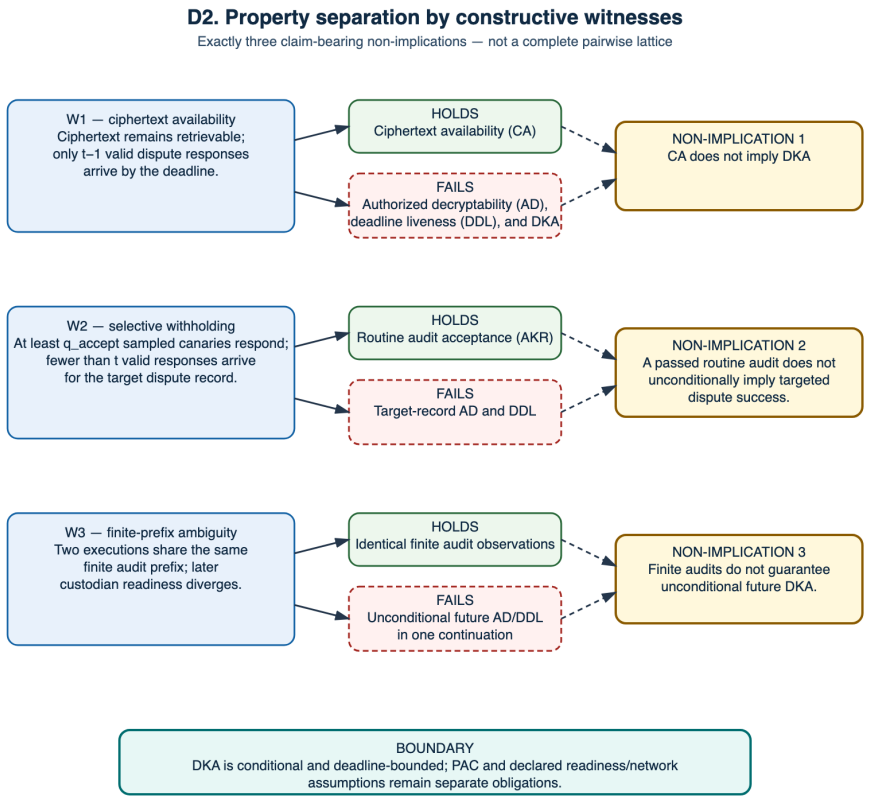


Figure 2: D2. Three constructive property-separation witnesses, each routing a held condition and a failed condition to one bounded non-implication.

D2 — Constructive property separation. Exactly three witnesses support the draft analytic obligations in C001: ciphertext availability does not imply deadline-bounded DKA; routine audit acceptance does not unconditionally imply targeted dispute success; and a finite audit prefix does not guarantee future DKA. This is not a complete pairwise property lattice.

D3 — Readiness-audit sequence. The routine path uses fresh, context-bound canaries and classifies valid, invalid, and missing responses at the declared deadline. It does not exercise production ciphertext.

D4 — Authorized dispute sequence. The diagram records the intended policy and threshold-release ordering; it does not establish deployed confidential transport, deadline success, or production authorization.

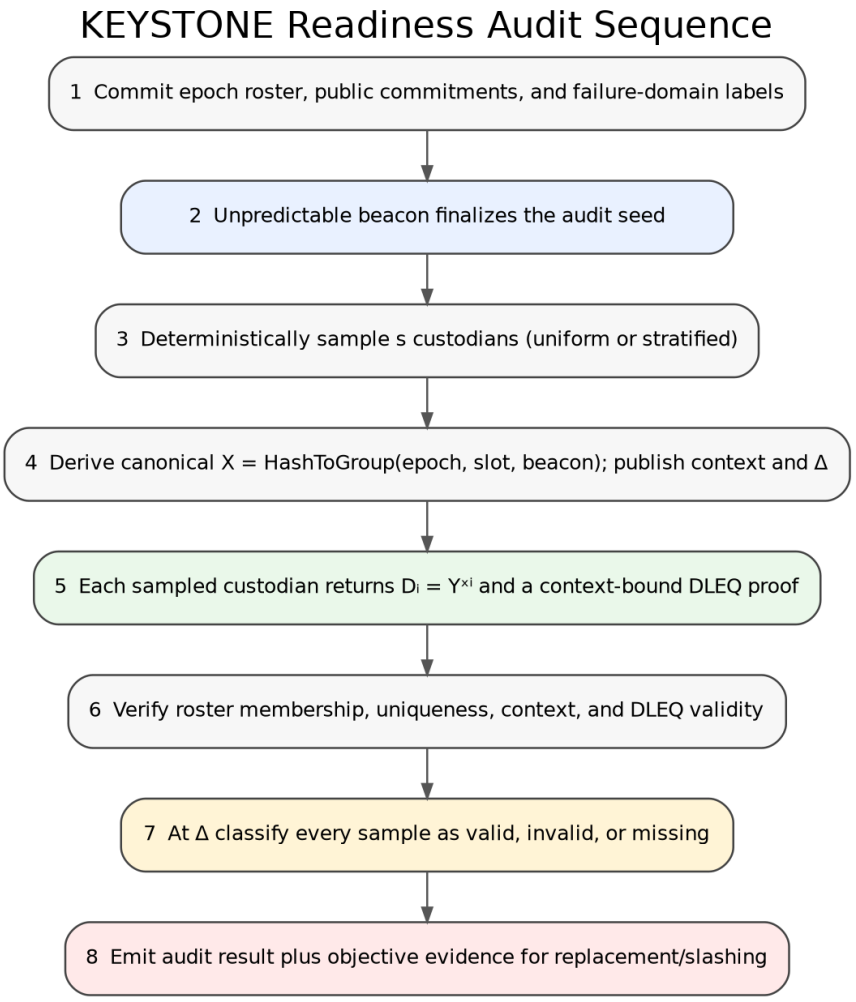


Figure 3: D3. Eight-step readiness-audit sequence from committed epoch data and beacon sampling through canary proof verification and evidence emission.

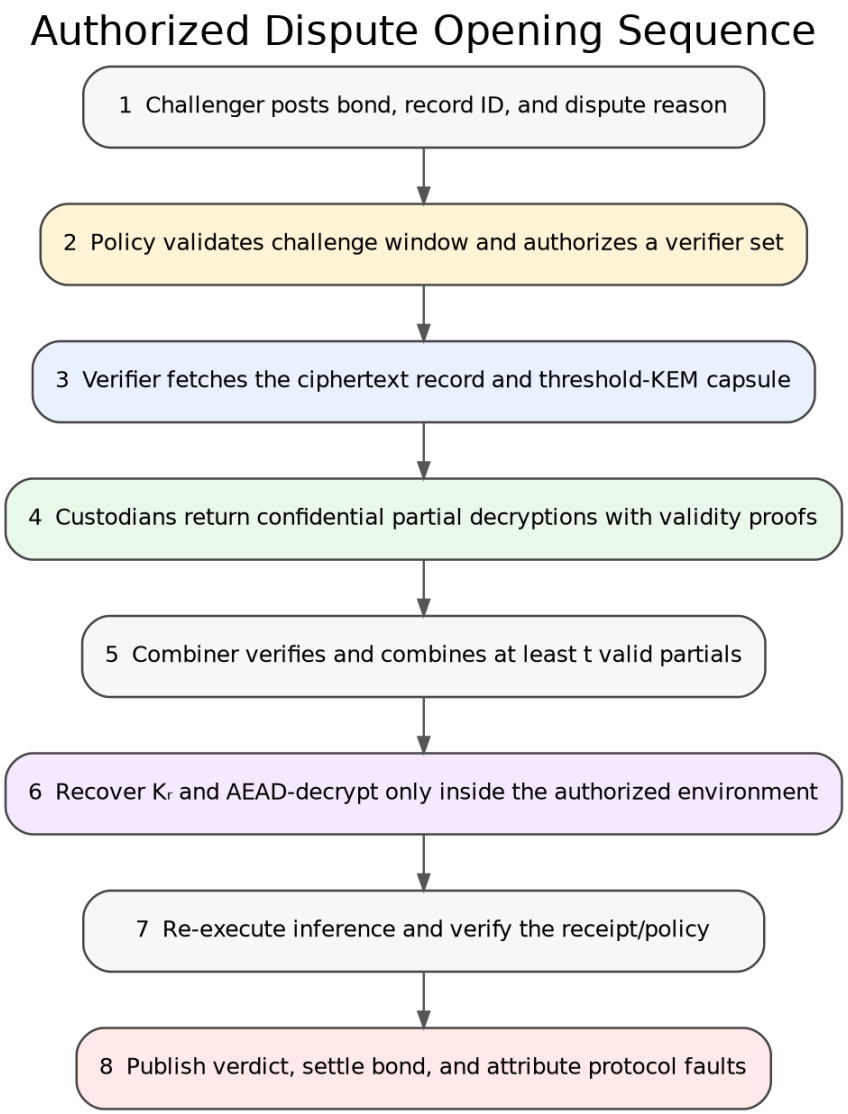


Figure 4: D4. Eight-step authorized-dispute sequence from challenge validation through confidential partial release, threshold combination, re-execution, and verdict.

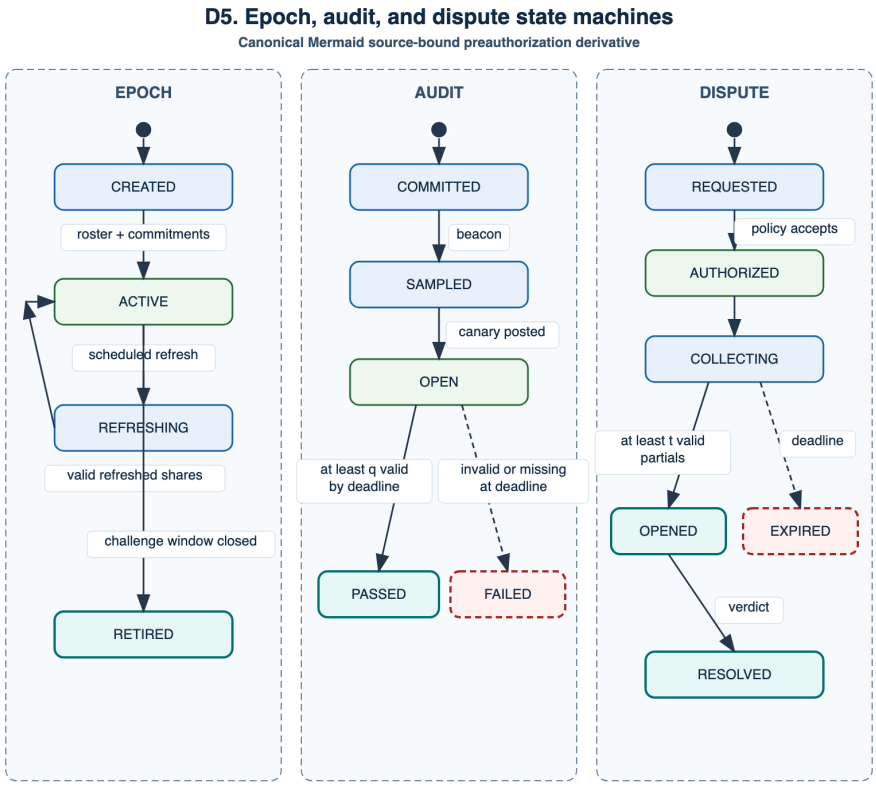


Figure 5: D5. Epoch, audit, and dispute state machines with explicit successful, failed, expired, and retired terminal paths.

D5 — State machines. Allowed transitions make terminal and failure paths explicit for implementation review. The current contract is still a non-production boundary with admin-adjudicated off-chain proof outcomes.

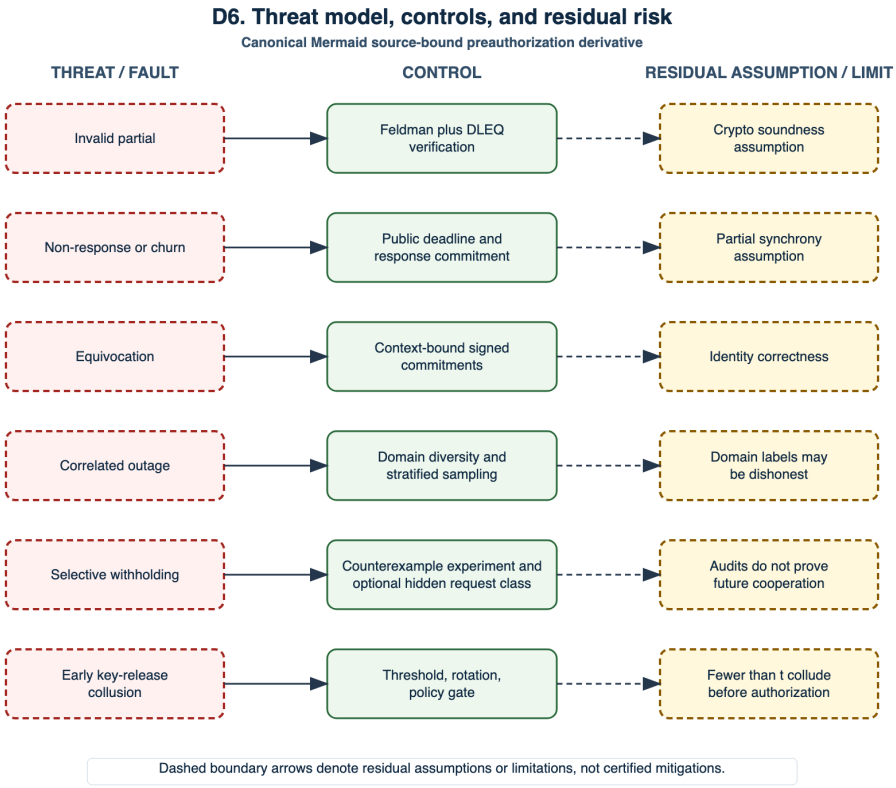


Figure 6: D6. Six adversary or fault classes mapped through KEYSTONE controls to six residual assumptions or risks.

D6 — Threat model and residual risk. Each control is paired with a residual assumption. In particular, domain labels may be dishonest, audits do not prove future cooperation, and the pre-authorization collusion threshold remains an assumption rather than an observed result.

D7 — Sampling and failure domains. This is a method illustration under truthful domain labels. It does not claim universal superiority of stratification or field prevalence of the illustrated outage.

D8 — Experiment-to-manuscript pipeline. The flow separates configuration, execution, versioned outputs, displays, claim binding, and review gates. A clean local pipeline does not substitute for authorized confirmatory analysis, independent reproduction, or human submission approval.

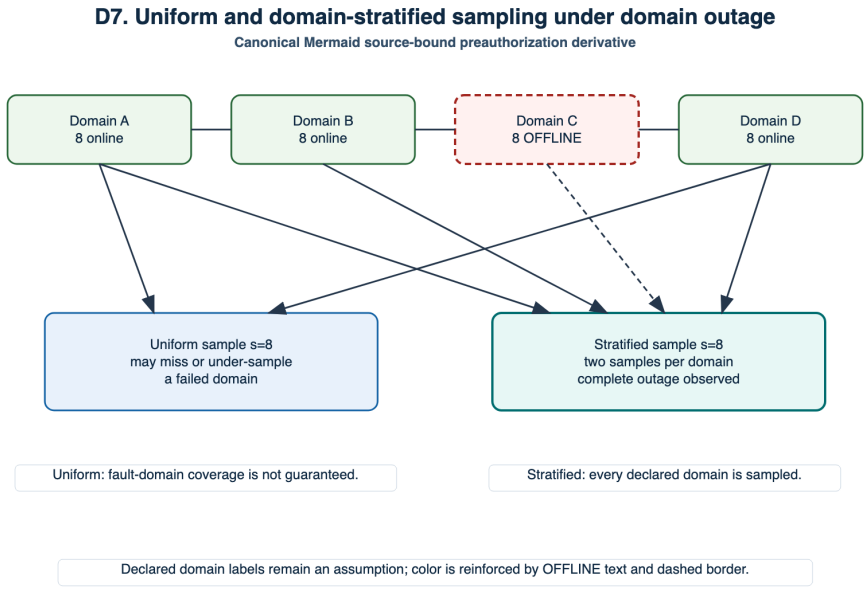


Figure 7: D7. Four fault domains with one offline domain and contrasting uniform versus domain-stratified sample routes.

Results

The present results surface is an explicitly preliminary inventory. The design-layer `03-design/primary-results-contract.csv` exists only as a schema-complete, non-executable result contract: all nine slots remain unauthorized, non-independent, `NOT_COLLECTED`, and `NOT_ESTIMATED`. It does not replace missing authorized confirmatory evidence.

The temporary Markdown/LaTeX preliminary-table bundle has a fail-closed same-workspace lineage receipt that binds its five input files, exporter, selected fields and row identities, section mapping, and both outputs. This improves draft reproducibility only; it is not a canonical T-series result, scientific verification, or independent reproduction.

A second-agent, same-host isolated-copy mechanical rerun subsequently executed the frozen ten-command package check under tool-level offline flags. The first fresh attempt failed closed when Foundry rejected a non-boolean offline setting; its hash-bound failure receipt and logs were preserved. After the environment contract was corrected and independently re-reviewed, a new operator completed all ten commands with zero return codes, unchanged source and isolated-copy inventories, and a hash-verified `MECHANICAL_PASS` receipt for that historical snapshot. The canonical reproducibility report at `05-analysis/reproducibility-report.md` finds that the receipt's source-

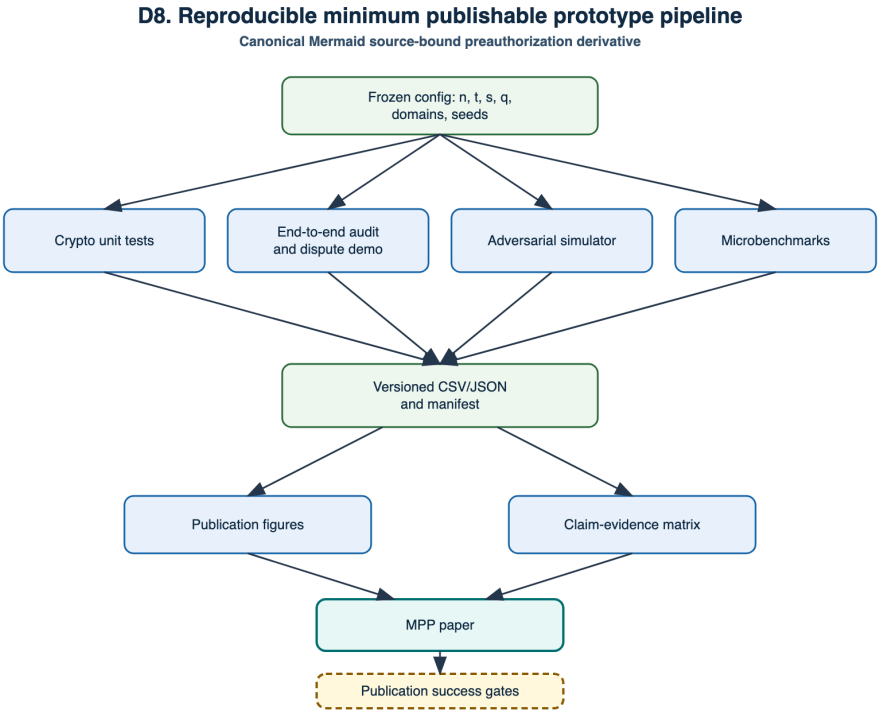


Figure 8: D8. Reproducibility pipeline from frozen configuration through tests, simulation, data products, figures, claim mapping, manuscript assembly, and publication gates.

manifest and program-state hashes do not match the current revisions, so a current-tree replay is required. Even after such a replay, this establishes bounded same-host mechanical consistency only: it used the same machine and potentially the same tool caches, did not enforce network isolation at the kernel level, did not execute the distributed deadline study, and is neither external reproduction nor independent scientific verification, institutional approval, venue approval, or evidence that changes the current claim-maturity ceilings.

7. Evaluation

Only a limited set of result classes is admissible for drafting:

- analytic static catastrophic false-accept and detection calculations;
- deterministic correctness and interoperability fixtures;
- internal preliminary IID, correlated-domain, stratified-sampling, and selective-withholding simulation outputs;
- internal local cryptographic timings and contract test-body gas measurements.

The current figures F1–F5 and `paper/tables/t1_t8_package.*` are draft preauthorization display assets. They are not canonical confirmatory tables or publication evidence. Missing or blocked evaluation outputs include authorized primary outcomes, confirmatory robustness and negative-result analyses, deadline Figures F6–F7, the planned recovery-timeline Figure F8, independent reproduction, and external validation for `C003`. The canonical `05-analysis/results/exploratory-findings.csv` now records the four temporal Markov rows under distinct `EXPLORE-*` identifiers. It preserves the simulated source maturity but caps claim support at `V0 ASSERTED`: each row is explicitly post-hoc, preauthorization-only, unauthorized, non-independent, and descriptive rather than confirmatory. Its conditional denominator is the number of trials ending in the catastrophic state, not all simulated trials. Draft `04-data` provenance and evidence-status ledgers hash-bind the source inputs; the deadline result remains explicitly `BLOCKED` with no executed provenance row. These ledgers improve traceability only and do not activate scientific support edges or advance the canonical `INTAKE` phase.

The canonical draft `05-analysis/results/negative-findings.csv` now records only the interval-separated selective-withholding gap already present in the frozen simulation (`NEG-SW-11` through `NEG-SW-14`). Each row remains a preauthorization simulated negative finding, unauthorized, non-independent, and capped at `V0 ASSERTED`. This closes a traceability gap but is not an authorized confirmatory result and does not provide deadline, production, external, field, adaptive-adversary, or universal-security evidence.

The canonical draft `05-analysis/results/robustness-and-boundaries.csv` now preserves all 17 frozen IID outage cells and all 12 frozen matched-seed sampling policy cells as a deterministic traceability ledger. Every row is explicitly `DRAFT_ROBUSTNESS_BOUNDARY_ONLY`, preauthorization, simulated, unautho-

rized, non-independent, and capped at `V0 ASSERTED`. Correlated-domain output is excluded because its current result-ID binding is entangled with the quarantined Markov asset; Markov, selective-withholding, and deadline rows are also excluded from this ledger. The file therefore does not establish confirmatory robustness, policy optimality, production behavior, deadline performance, generality, or external validity.

The future confirmatory outputs are predeclared in `02-feasibility/pilot-run-contract.csv`. The minimum core comprises the separation witness, exact static result, deterministic crypto and contract boundaries, and the limitation-bearing selective-withholding family. IID and stratification are supporting qualification families; correlated-domain analysis is excluded pending a truthful domain-label source. The included draft counts and stream schedule are recorded in `03-design/pc03-prospective-counts.csv` and `03-design/pc03-seed-schedule.csv`; these are design inputs, not results. The distributed deadline result remains an extended full-paper family and must be excluded, rather than imputed, if its environment and precision gates remain open.

7.1 Preliminary quantitative display map

The five available quantitative figures are included below only to freeze their drafting roles and captions. Their source data, generator, SVG/PNG hashes, and automated visual-QA status are recorded in the source and figure manifests. They remain preliminary and must be replaced or re-authorized from canonical analysis ledgers before submission.

F1 — Static detection bound. Analytic one-audit probability under the declared fixed catastrophic state and uniform-sampling assumptions; not a temporal availability or field-frequency result.

F2 — IID failure sweep. Internal preliminary simulation with the configured scenario grid and uncertainty calculation; not confirmatory or externally validated evidence.

F3 — Correlated-domain behavior. Internal model-bounded simulation; domain identity correctness and real outage structure remain unverified assumptions.

F4 — Selective-withholding limitation. Preserved negative result that blocks an unconditional audit-to-dispute-success inference.

F5 — Sampling-strategy comparison. A matched, model-specific internal comparison. It does not establish universal dominance, external validity, or production benefit.

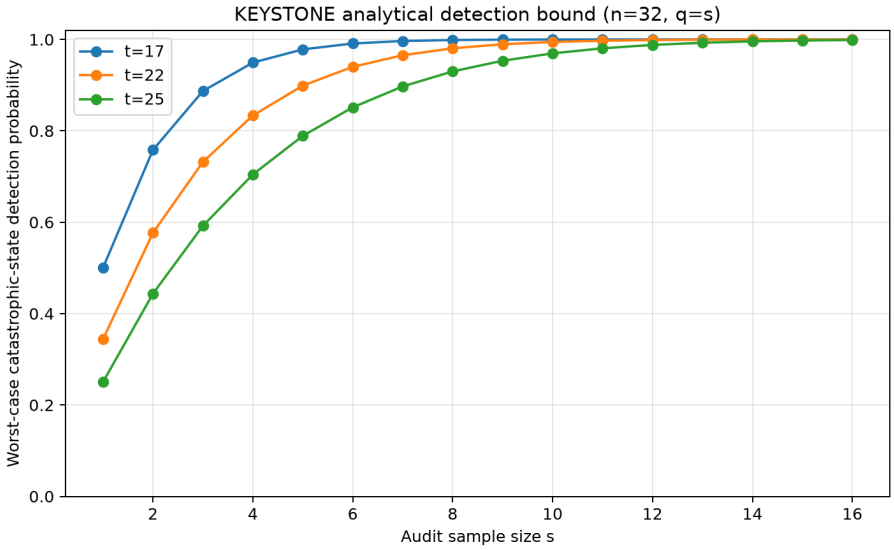


Figure 9: F1. Static one-audit detection bound under uniform sampling as the sample size changes.

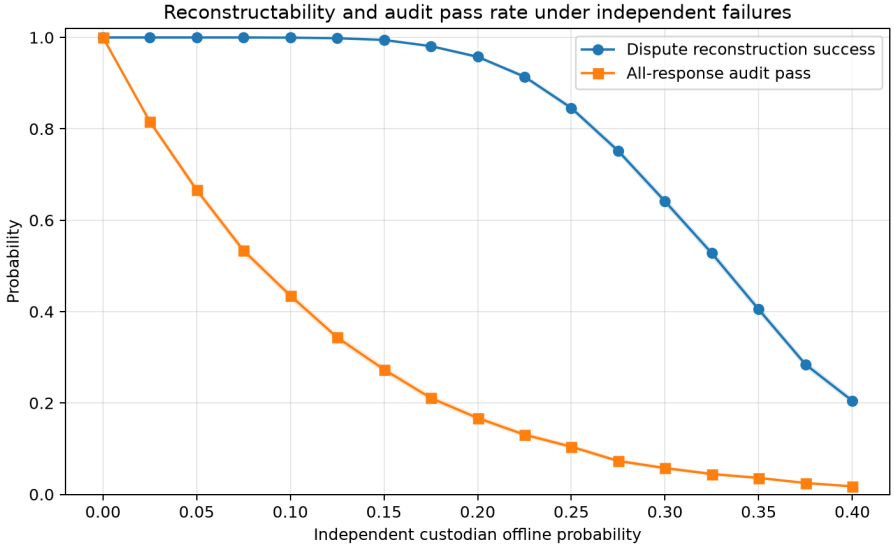


Figure 10: F2. Internal IID outage sweep showing reconstruction and audit-pass estimates with Wilson intervals.

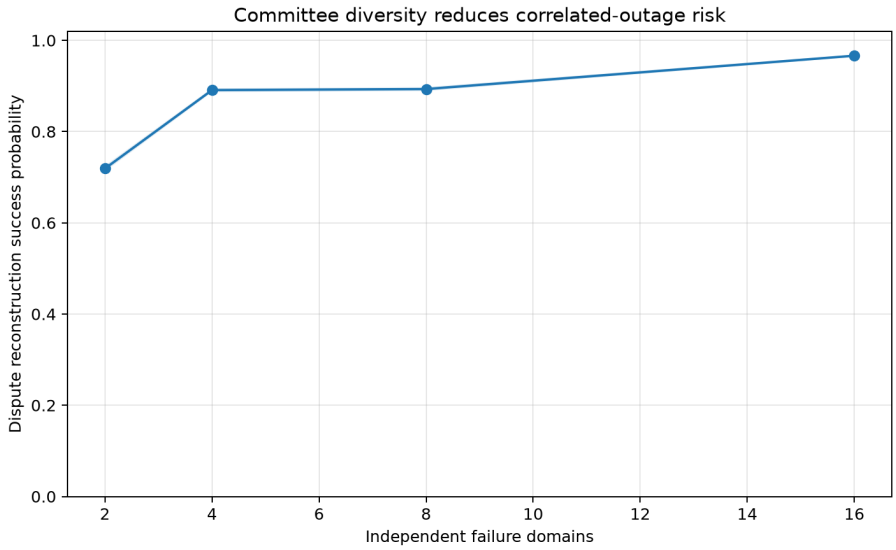


Figure 11: F3. Internal correlated-domain outage sweep under declared truthful domain labels.

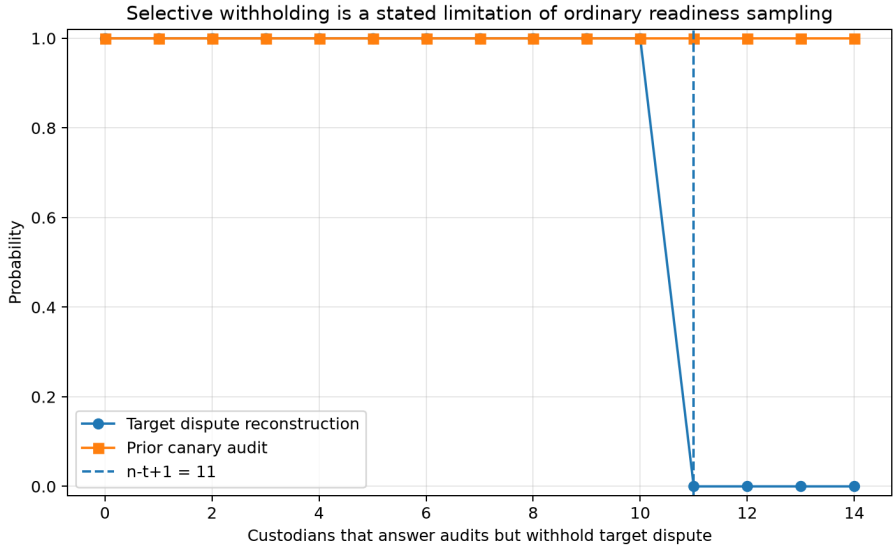


Figure 12: F4. Selective-withholding negative-result figure showing that routine audit acceptance can coexist with failed targeted dispute reconstruction.

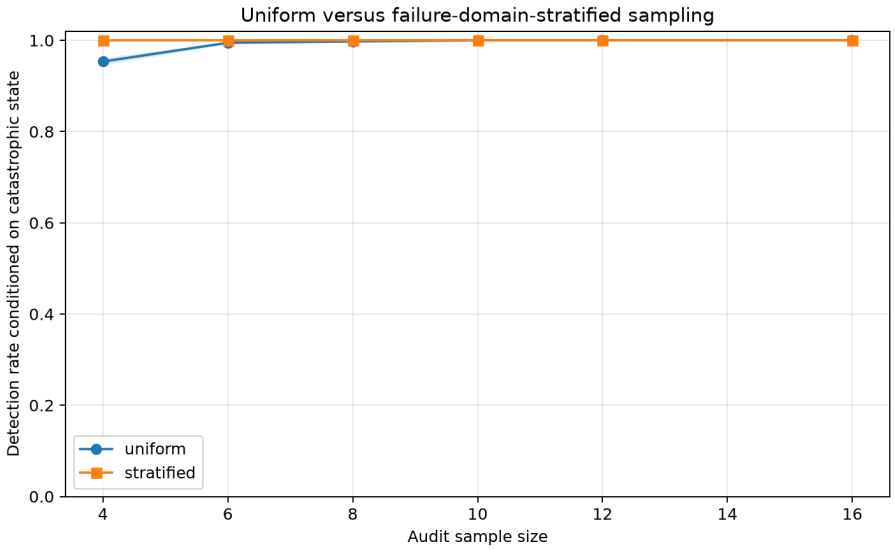


Figure 13: F5. Matched internal comparison of uniform and domain-stratified sampling under the configured conditioned catastrophic states.

7.2 Draft evidence-bound table map

The editable T1–T8 package is now available in `paper/tables/t1_t8_package.md` and `paper/tables/t1_t8_package.tex`. Every table remains draft / pre-authorization only and inherits the canonical ceiling of the underlying claim or evidence source.

- `T1` bounds the strongest-prior-art matrix and does not clear novelty.
- `T2` is a design-comparator registry only and makes no measured-superiority claim.
- `T3` records frozen local conditions and the still-missing `RID-C003-DEADLINE-001` distributed profile.
- `T4` displays preauthorization simulated results parsed from canonical JSON outputs only, capped at `V0 ASSERTED` for manuscript claims.
- `T5` is a planned mechanism-isolation registry; every row remains `MISSING_NOT_EXECUTED`.
- `T6` presents the canonical robustness ledger plus separately classified exact and exploratory temporal checks; selective-withholding negative findings remain in `T8`.
- `T7` limits feasibility evidence to local timings and Foundry gas while leaving distributed, operator, network, and external rows blocked.
- `T8` preserves selective withholding, synchrony limits, truthful-domain-label dependency, and missing external validation as explicit negative or unresolved findings.

8. Discussion

The current safe interpretation is that ciphertext availability and dispute-key serviceability should not be treated as interchangeable; routine readiness audits may characterize present serviceability only under stated assumptions; failure-domain modeling and sampling semantics affect the meaning of reported probabilities; and local prototype success does not establish external reproducibility, field readiness, or production security.

Stronger interpretation must await reconciled prior art, authorized analysis, and claim-matched independent evidence.

9. Limitations and Future Work

- Novelty remains `UNRESOLVED`; the broad historical claim is not submission-safe.
- The canonical feasibility decision remains `UNASSESSED`; the draft feasibility package recommends a blocked posture and only a conditional `PILOT_FIRST` path after prerequisites clear.
- Current stochastic outputs are internal and mostly pre-authorization.
- Temporal dependence is exploratory; selective withholding is a preserved negative result.
- The distributed deadline benchmark and independent external validation are absent.
- Independent threshold-cryptography and distributed-systems reviews are absent.
- No venue portfolio, current reporting checklist, hermetic PDF package, rendered-page review, or accountable human approval exists.

10. Conclusion

`PENDING - EVIDENCE-BOUND SYNTHESIS.` The conclusion may restate the distinction between ciphertext availability and dispute-key serviceability and the conditional nature of any audit assurance. It must not imply future-proof availability, external validity, production readiness, or publication readiness unless the corresponding canonical gates later pass.

Data and Code Availability

Current local source assets include `prototype/src/keystone/*`, `prototype/tests/*`, `prototype/results/*`, `contracts/src/*`, `contracts/test/*`, `contracts/gas_report.csv`, `paper/test_vectors.json`, `paper/signature_test_vectors.json`, `diagrams/*`, and the two `*/results/engineering_qa/*-preauth-receipt.json` bundles. Availability does not imply evidence authorization or final reproducibility. The manuscript assembly inventory is tracked in `07-manuscript/source-manifest.json`. Draft source-to-result lineage is recorded in `04-data/provenance-manifest.csv` and `04-data/evidence-status.csv`; those files retain the preauthorization

and same-workspace limitations of their inputs. The derived temporal ledger at `05-analysis/results/exploratory-findings.csv` is reproducible from the hash-bound Markov dataset through `scripts/export_exploratory_findings.py`, but is quarantined from confirmatory, novelty, feasibility, and external-validation claims. The derived robustness ledger at `05-analysis/results/robustness-and-boundaries.csv` is reproducible from the hash-bound IID and sampling-policy datasets through `scripts/export_robustness_boundaries.py`; it remains a draft traceability surface and cannot substitute for authorized primary evidence, confirmatory analysis, independent reproduction, or external validation. The schema-complete primary-results contract at `03-design/primary-results-contract.csv` contains no observations or estimates; the canonical `05-analysis/results/primary-results.csv` remains missing; the current reproducibility disposition and historical-receipt binding audit are recorded in `05-analysis/reproducibility-report.md`. The non-executable future-run contract is recorded in `02-feasibility/pilot-plan.md` and `02-feasibility/pilot-run-contract.csv`; it contains no collected outcome and grants no authorization. The PC03 amendment, count contract, and seed schedule under `03-design/` are likewise result-blind, non-executable planning artifacts pending independent methods verification.

Declarations and Pending Human Inputs

Draft author metadata inputs have been supplied for prospective authors, and the accountable human reported no institution, employer, funding agreement, NDA, venue policy, or export-control conflict with the current charter or study profile. At the author's request, final author order, corresponding-author designation, accountable metadata freeze, and exact affiliation wording remain deferred. Funding, conflicts of interest, CRediT roles, data/code release terms, AI-use disclosure, and venue-specific institutional checks also remain pending. No AI-generated text in this shell constitutes author approval or submission authorization.

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